A fall detection method for intelligent healthcare

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***Abstract*—Falls are a prevalent health issue and one of the leading causes of serious injuries among elderly individuals. Detecting falls as quickly as possible is essential to minimize fall-related damage. Although sensor-based fall detection approaches provide high detection rates, wearable sensors can be inconvenient for users to wear consistently, and large space sensors are prone to noise interference and low accuracy. In this paper, we propose a vision-based fall detection approach that utilizes human poses and a long short-term memory (LSTM) network. We extract skeleton data from videos and stack them into sequences, which we use as input features for the LSTM network to classify fall activities. By using skeleton data, we can mitigate the effects of environmental factors such as lighting variations and complex backgrounds. As a result, our proposed fall detection method achieves a high accuracy of 93% on the URFD and 97% on the CAUCA Fall public datasets.**

Keywords—fall detection, intelligent healthcare, pose estimation, YOLOv7, long short-term memory.

I. INTRODUCTION

Modern advances in medicine have increased life expectancy, resulting in a global phenomenon of population aging. According to the World Health Organization (WHO), by 2030, one in six people worldwide will be aged 60 years or older [1]. As a result, the need for healthcare for the elderly is becoming increasingly important. Statistics from the Centers for Disease Control and Prevention (CDC) indicate that in the United States, approximately 28% of adults aged 65 and older report falling each year, resulting in an estimated 8 million fall injuries that require medical treatment or restrict activity for at least one day [2]. Therefore, to provide timely medical assistance and minimize the injuries caused by falls, fall detection (FD) alarm systems for older people are being researched and developed worldwide.

Although the general method involves using sensors to detect sudden changes in people's movement during falls, there are many differences in technology and engineering between these approaches. These approaches can be divided into three categories: wearable device-based, ambient-based, and vision-based [3]. Wearable device-based systems use accelerometer sensors mounted on the body to detect falls. Many commercialized smart devices such as wristbands, wristwatches, and necklaces have integrated this technology. However, the limitations of this method include the inconvenience of charging or wearing and single-person fall detection. Ambient-based FD systems use light, motion, vibration, and proximity sensors to detect falls in a large space, but the accuracy is influenced by the surrounding environment. Vision-based approaches offer many advantages over sensor-based systems. Recent developments in computer vision and machine learning have improved the accuracy of vision-based FD systems. The challenges of vision-based FD include decreasing accuracy as lighting conditions change, complex backgrounds, many daily activities that look like falls, and a lack of data to train the model.

Our proposed methodology comprises two essential stages. Initially, we utilize a precise model to extract human postures and compile them into sequences. After conducting various pose estimation models, we selected YOLOv7-pose [4, 5] to extract key points from input videos. These key points contain important characteristics of human fall activities. In the second stage, these extracted features are fed through a classifier that distinguishes between "Fall" or "Normal" action sequences. To handle temporal information in the video classification task, we decide to use the LSTM model as a classifier.

In summary, our proposed method provides the following main contributions:

- Our proposed methodology represents a significant advancement in the field of fall detection by combining human posture data to reduce the impact of environmental factors. Experimentation has revealed that the use of pose estimation based on YOLOv7 performs superior results compared to alternative techniques.
- Combine with object tracking to detect falls for all individuals in the frame. We use a simple object-tracking technique based on changes in the center of objects, which ensures that the entire system processes data quickly and efficiently.

The remainder of this article is organized as follows. Section II presents the related work. Section III outlines our proposed method, while Section IV presents the results and evaluation.

II. RELATED WORK

Fall detection systems can be categorized into three types: ambient-based, wearable device-based, and vision-based. Various approaches for fall detection have been proposed, each with its advantages and disadvantages. Ambient-based methods use sensors such as light, motion, vibration, and proximity sensors to collect daily activity data and detect falls. This approach is advantageous due to its low cost and user convenience. However, the sensors' accuracy may be imprecise in certain situations, such as changes in lighting or concussions, due to their reliance on the surrounding environment. In contrast, wearable sensors, as suggested by authors [6], use two accelerometers and gyroscopes mounted on the chest and thighs. The sensor signals can be processed to determine the body's angular velocity and acceleration to detect actions that lead to falls using threshold levels. While this method is highly accurate, its limitations include the potential for misidentification of actions such as jumping on the bed or falling against the wall, and discomfort for the user due to the need for battery charging.

Today, computer vision and machine learning have enabled vision-based fall detection approaches to achieve better results. For instance, based on the rarity of falls, researchers [7] have used a 3-D convolutional network and AutoEncoder to detect falls. The main concept is to allow the neural network to learn how to reconstruct an entire video sequence of normal activities. When an abnormal action such as a fall occurs, there is a significant difference between the reconstructed video and the initial video. The advantage of this approach is that exact labeling of data is not necessary; the model only requires normal daily activities data, which typically constitutes the majority, to train. However, the limitations of this method include the potential for misidentification of other abnormal actions, and the processing time and complex computation may not be suitable for real-time applications.

Research [8] uses Histogram of Oriented Gradients (HOG) algorithm to descript shape of human. Human body shape changes are calculated and combined with the SVM classifier for fall detection. The limitation of this method is that the accuracy of the model depends on the difference between the color of the subject and the background. Research [9] utilizes optical flow algorithms to convert a time-series image into a single image that captures all movements over time. The generated image is passed through a convolutional neural network to determine whether the input time-series data is classified as "Fall" or "Normal". The results can produce high accuracy, but the disadvantage of this method is that the optical flow algorithm only detects movements, even with objects such as clothes and handbags. This limitation can be remedied by recently developed posture recognition algorithms.

Research [10] has combined human posture recognition models and SVM algorithms. Specifically, the posture recognition model returns the position of the body's key points, which are classified as falls based on the SVM algorithm. In addition to using extracted key points, research [11] calculates other characteristics of the fall, such as body velocity, the angle between the body and the ground, and the ratio of the "bounding box" around the body for classification based on threshold levels. The advantage of using posture recognition algorithms is to focus only on human objects, thereby increasing the accuracy of the model. However, the limitation of studies [10] and [11] is that they only use one frame for identification, which can confuse with images such as lying and crouching.

Research [12] uses the pose estimation model to extract the position of key points, after which this data is processed through the LSTM network to classify the types of actions. The model is built from the MobilenetV2 network to increase processing speed and can classify seven indoor postures, including lying, sitting, crouching, standing, walking, fighting, and falling. The model can achieve an accuracy of up to 99%, but the accuracy in fall detection is 83%. Research has also shown the potential of time series classification in vision-based fall detection.

These studies have inspired us to develop a fall detection system based on human pose estimation models that combine body geometry features and LSTM classifiers.

III. PROPOSED METHOD

A. *System Overview*

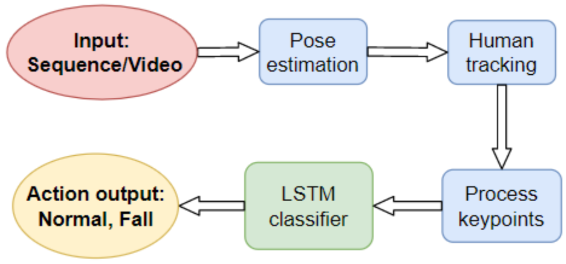


Figure 1. System overview

The block diagram of our proposed system is shown in Fig. 1. The system works as follows:

- The input is a continuous sequence of RGB images performing fall activities or other daily activities. These frames contain the body of one person for training and many people for testing.
- Skeletons data was extracted by using the YOLOv7-pose algorithm.
- The person tracking block follows people to stack their skeletons in the testing phase.
- The key points of each person are stacked into sequences and normalized.
- These sequences are passed through the LSTM classifier for training and testing.

The output of the system is human skeletons and bounding boxes with classification labels **fall** or **normal.**

B. *Dataset and preprocessing*

Three datasets are used to train and evaluate the model:

- UR Fall Detection dataset (URFD) [13] comprises 70 action sequences with different subjects and backgrounds: 30 videos of falling actions and 40 videos of daily activities close to falls, such as sitting, lying, and bending low.
- CAUCA Fall Dataset [14] contains 100 videos with occlusions, different brightness levels, participants, movements, backgrounds, and distances from the camera to the fall. There are 50 falls and 50 non-fall activities recorded.
- Additional data, including fall videos on YouTube and other sources, were collected to increase the diversity of our training dataset.

The collected data comprises various contexts, subjects, and shooting angles. Fig. 2 shows an example for each dataset. To increase the amount of training data, we divided each video into sets of 60 consecutive frames corresponding to action sequences lasting between 2-3 seconds.

(a)

(b)

(c)

Figure 2. An example of datasets (a) CAUCA fall dataset (b) UR fall dataset (c) Our collected fall data dataset

C. *Human pose estimation*

Posture recognition is performed by YOLOv7-pose, an open-source model for multi-person pose estimation. YOLOv7-pose was developed from YOLO-pose [4] and YOLOv7[5] framework for human pose estimation. YOLOv7-pose is a single-stage multi-person key points detector that allows simultaneous detection of both bounding boxes and key points of multiple people. The input is RGB images. Instead of focusing on 80 classes for classification, the output of YOLOv7-pose includes the position and confidence of both bounding box key points. Keypoints confidence represents the visibility of key points. This value will be closer to 1 if that key point is visible or occluded and closer to zero if that key point is outside the field of view. With 17 key points prediction, each anchor has a prediction vector defined as follows:

$$P_v = \{C_x, C_y, W, H, Conf_{box}, Conf_{class}, K_x^1, K_y^1, K_{conf}^1, \ldots , \ldots K_x^{17}, K_y^{17}, K_{conf}^{17}\} \quad (1)$$

YOLOv7's improvement in speed and accuracy helps YOLOv7-pose achieve better results than multi-person pose estimation models.

The comparison of YOLOv7-pose with OpenPose[15] is shown in Fig. 3. YOLOv7-pose can capture all key points correctly during the fall with a suitable confidence score.

D. *Data processing*

The raw data of each person, after being tracked, is processed before being passed to the classifier. This process consists of three main steps:

Step 1 (filtering): The output of YOLOv7 has 57 parameters, including the bounding box, coordinates of key points, and confidence scores. Some key points may be lost during processing. However, because YOLOv7's pose estimation has high accuracy, the lost key points problem can be handled by choosing lower thresholds for key point confidence. When there are enough key points, the necessary data frame only includes the coordinates and positions of the bounding box and key point.

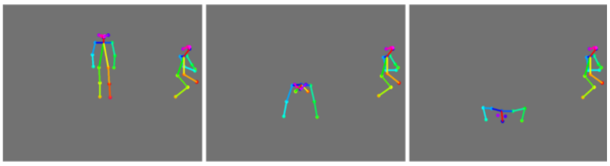
Step 2 (scaling center): The body's initial position can be anywhere in the frame. To obtain the best quality features, we transform the data frame above to the center position for all detected objects.

Step 3 (normalization): The position of points and bounding box is scaled to the height and width of the image. To increase the accuracy of the model and reduce the convergence speed of the model, we normalize the data to the interval [-1, 1].

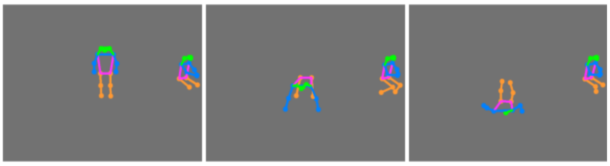
E. *Classifier and training*

Falls are a continuous sequence of actions that require both spatial and temporal information for the recognition task. As shown in Fig. 4, we use a many-to-one LSTM network to detect fall activities.

(a)



(b)



(c)

Figure 3. Comparison of YOLOv7-Pose and other pose estimation models (a) Input images (b) The output of OpenPose (c) The output of YOLOv7-Pose

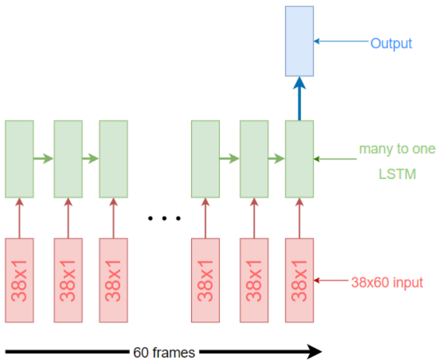
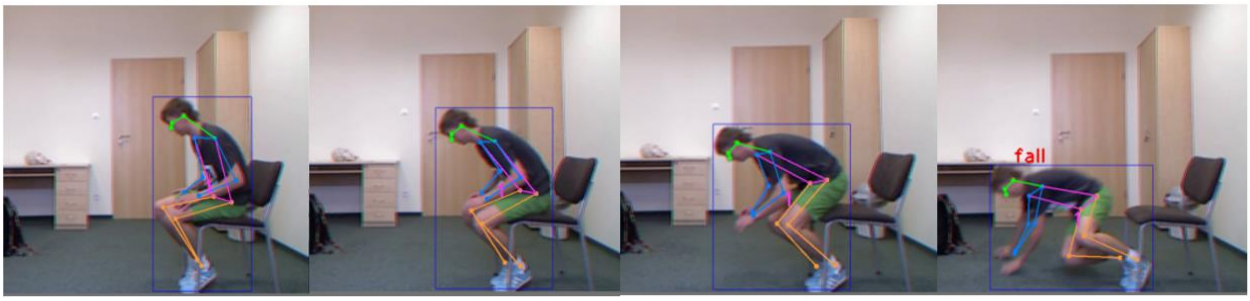


Figure 4. Many to LSTM model for fall detection.

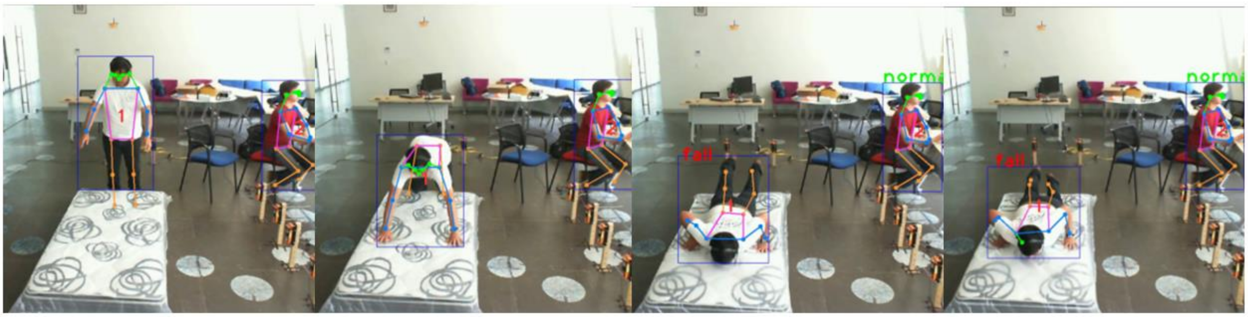
LSTM is developed from Recurrent Neural Network (RNN) - a neural network that can process sequential and time series data such as text, audio, and video. RNN carries some amount of information from previous states to the next stage and finally combines them to predict the result. However, carrying too much information can cause a vanishing gradient, leading to information loss. To avoid long-term dependency problems, LSTM is designed by adding more compute gates to control how much information is hidden in each state and what information to output.

Our proposed LSTM network has an input of 60 features extracted from consecutive frames. After removing confidence scores from YOLO-pose outputs, each feature has 38 parameters. We labeled each 60-feature set with either 0 or 1, representing non-fall and fall results, respectively.



(a)

(b)



(c)

Figure 5. Results of fall detection on test datasets (a) Result on URFD dataset (b) Result on CAUCA dataset (c) Multi-persons fall detection

IV. RESULT AND EVALUATION

The outputs of our proposed method contain positions and predictions (fall/normal) for each person in the frames. Fig. 5 shows the model results with corresponding input sequences.

Like other classification models, we use a confusion matrix shown in Tab.1 to evaluate and focus on precision, recall, and accuracy. We also use sensitivity and specificity to compare with other methods.

The evaluation criteria are calculated as follows:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall\ (Sensitivity) = \frac{TP}{TP + FN}$$

Table 1. Confusion matrix of fall detection

Actual / Predict	Fall	Normal
Fall	True Positive (TP)	False Positive (FP)
Normal	False Negative (FN)	True Negative (TN)

$$Specificity = \frac{TN}{TN + FP}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Table 2. Results of our method on test datasets

Dataset	Precision (%)	Recall (%)	Accuracy (%)
URFD	90.3	93.3	93.0
CAUCA	96.0	98.0	97.0

The results of our proposed method evaluation on two separate datasets, URFD and CAUCA, are presented in Tab.2. Each video in the aforementioned datasets portrays the activity of an object and is captured from various camera angles. The assessment of the model on these datasets, which were produced by a multitude of participants with varying physical attributes, emphasizes its capacity to perform optimally across a diverse array of real-life scenarios.

The data reveals that the model performs better in the CAUCA dataset compared to the URFD dataset. One of the primary reasons for this distinction can be attributed to the camera angle during image capture and the primary objective of each dataset.

The CAUCA dataset, which was constructed in 2022 with the primary purpose of addressing the fall prediction problem through computer vision, was recorded using an oblique camera angle that offered superior image quality compared to the URFD dataset captured in 2014. The actions in the CAUCA dataset were comprehensively recorded through a side view camera mounted at the top of the room, as depicted in Figure 5b, leading to a model with excellent predictability.

On the other hand, the URFD dataset was recorded in 2014 to be used for various methods such as vision-based and sensor-based. The dataset was captured using both RGB and depth cameras, as well as sensor data. The camera angle in the URFD dataset was placed horizontally relative to the object's motion, making it challenging to capture the movement accurately, particularly for actions such as falling forward or backward. As a result, the predictive model's performance was inferior to that of the CAUCA dataset.

Table 3. Compare of our results with other method on URFD dataset

Related work	Method	Specificity (%)	Sensitivity (%)
Bogdan Kwolek [13]	Device-based + lower threshold	80.0	93.3
Yixiao Yun [8]	HOG + SVM	89.7	96.8
Yangsen Chen[10]	OpenPose + SVM	95.8	95.8
Ours	**YOLOv7-pose+LSTM**	**94.9**	**93.3**

Due to the recent introduction of the CAUCA dataset in 2022, few studies have utilized it thus far. In order to compare the efficacy of our model to other approaches, we evaluated two metrics, namely Specificity and Sensitivity (Recall), using the URFD dataset. Although our proposed method did not demonstrate superior performance in comparison to prior studies, it proved capable of being applied and estimated on frames featuring multiple individuals.

With regards to identifying falls in frames with multiple persons, our methodology involved tracking the movement of each individual in several consecutive frames and generating predictions for each person within the frame. As a result, the method's outcomes not only predicted the falling behavior of the individual within the frame but also identified their falling position. The simulation outcomes are presented in Figure 5c. The model's ability to operate on video frames that feature multiple individuals emphasizes its practical applicability, which has yet to be achieved by the studies presented in Tab.3.

CONCLUSION

This paper proposes a vision-based fall detection method that relies on human skeletons. Our approach employs YOLO-Pose for pose estimation and LSTM to recognize fall activities. The system is capable of functioning across a range of camera angles and achieves a high level of accuracy, with 93.3% accuracy on the URFD dataset and 97.0% accuracy on the CAUCA dataset. More specifically, our methodology can detect and localize fall activities in frames featuring multiple individuals, which underscores its practical applicability in real-world scenarios.

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